

Research paper

Association between cardiometabolic index and depression: National Health and Nutrition Examination Survey (NHANES) 2011–2014[☆]

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ABSTRACT

Background: Emerging evidence suggests a common pathophysiological basis for metabolic disorders and mental diseases. Despite the existence of reports suggesting a strong connection between dyslipidemia and depression, a comprehensive and reliable indicator to identify depression is still lacking. Cardiometabolic index (CMI) is an integrated index calculated from three vital metabolic indicators, including triglyceride (TG), high-density lipoprotein cholesterol (HDL-C) and waist height ratio (WHR).

Objective: This study aims to explore the association between CMI and depression.

Methods: Cross-sectional data of participants with complete information of CMI, depression, and other covariates were obtained from the National Health and Nutrition Examination Survey (NHANES). Weighted student's *t*-test and Chi-square test were used to identify the differences between two groups. Weighted multivariate logistic regression model, restricted cubic spline (RCS) regression analysis, subgroup analysis and interaction tests were conducted to explore the association between CMI and depression. Receiver operating curve (ROC) analysis and area under the curve (AUC) were also utilized to evaluate the performance of CMI in identifying depression.

Results: A positive correlation between CMI and depression was observed in 3794 participants included in the study, which was further confirmed to be non-linear via RCS regression analysis, with two significant inflection points being identified, including 0.9522 and 1.58. In the crude or adjusted models, individuals with a CMI level ≥ 0.9522 exhibited remarkably increased risk for developing depression. CMI got an AUC of 0.748 in identifying depression. Subgroup analyses and interaction tests indicate that the association between CMI and depression remained consistent across different subgroups and was not modified by other covariates except drinking. Those who are current drinkers and with a high CMI are more susceptible to suffer depression.

Conclusions: An elevated CMI is linked to increased risk for depression. Addressing dyslipidemia and improving lipid levels may potentially lower the risk for depression.

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1. Introduction

Depression is deemed as one of the most incapacitating psychological disorders and is correlated with significant individual, societal, and economic burdens (Bromet et al., 2011; Li et al., 2023). In light of the COVID-19 pandemic, there has been a significant surge in the global incidence of depression (Salari et al., 2020). It is undoubtedly that the depressive disorders have become a major health burden across different regions and ethnic groups. Currently, the management of depression encompasses various modalities such as psychotherapy, pharmacotherapy, electroconvulsive therapy and integrative therapy. Notwithstanding the broad and comparatively efficacious application of antidepressants in clinical settings, they frequently yield adverse effects and necessitate weeks before exhibiting effectiveness, thereby limiting their usefulness in specific populations, including those with acute conditions, adolescents, and pregnant women (Han et al., 2023; Jakobsen et al., 2020). Consequently, the prediction and prevention of depression is of paramount importance. Moreover, in current clinical practice, there is an urgent requirement for a reliable, accurate and easily accessible indicator to evaluate the onset and progression of depressive disorders.

The pathogenesis of depression is complex and can be modified by various factors, ranging from genetic, psychological, environmental to biological factors. In recent years, the relationship between metabolic profile with mental health has received increasing attention from researchers, and amounting evidences have revealed a common pathophysiological basis underlying the two conditions (Al-Khatib et al., 2022). The data from a meta-analysis suggested that individuals combined with an adverse metabolic profile (e.g., lipometabolic disturbance, glycometabolic disorder) obtained significantly higher risks for the development of depressive symptoms (Jokela et al., 2014). Previous research has indicated that individuals with depressive symptoms often exhibit specific changes in their lipid profiles, such as decreased levels of HDL, increased levels of LDL and triglycerides, and elevated levels of total cholesterol (TC) (Khalfan et al., 2023). Additionally, studies focusing on adults with major depressive disorder (MDD) have found that the severity of dyslipidemia and hypertriglyceridemia is significantly associated with the severity of depressive symptoms, including suicidal ideation (Oh and Kim, 2017; So et al., 2021). For instance, a study involving 601 adults with MDD found that a high ratio of triglycerides to HDL was positively correlated with an increased risk of developing MDD over a nine-year follow-up period (Watson et al., 2021). These findings strongly suggest a close association between lipometabolic disturbances and depressive disorders. Therefore, indicators that can effectively reflect changes in an individual's lipid profile may serve as useful biomarkers for depressive disorders.

The diagnosis of lipometabolic disturbances involves the integration of anthropometric and biochemical parameters. Therefore, some indicators for lipid metabolism profile have been established, such as visceral adiposity index (VAI) (Lazzer et al., 2023), which incorporates waist circumference (WC) and body mass index (BMI), and TG/HDL-C ratio (Kosmas et al., 2023), which integrates triglyceride (TG) and high-density lipoprotein (HDL), and the cardiometabolic index (CMI) (Lazzer et al., 2023; Liu et al., 2021), which effectively integrates the marker of adiposity (i.e., waist-to-height ratio/WtHR) with the TG/HDL-C ratio. Compared with other indicators described previously, the indexes of VAI and TG/HDL-C ratio have been shown to reflect lipid metabolism accurately in children and adolescents with severe obesity, as well as among obese adults (Lazzer et al., 2023). Moreover, amounting evidences have suggested that these lipometabolic indicators are closely linked to depressive disorders with favorable performance (Shi et al., 2021, pp. 2005–2018; So et al., 2021). However, as an integrated and comprehensive index based on anthropometric and biochemical parameters, whether CMI is closely connected with depression still remains unclear. Since the metabolic status of individuals with lipometabolic disturbance worsens over the years,

effective monitoring of lipometabolic disorders in depressive individuals is critical for developing an effective intervention to prevent the adverse health outcomes.

Hence, the objective of this study was to investigate the link between CMI and depression using the cross-sectional data from the NHANES database. The findings from our study could offer persuasive evidence regarding the role of lipid metabolism in depression and serve as a valuable reference for individuals and healthcare professionals in their daily practice.

2. Materials and methods

2.1. Data accession and study population

This research was conducted with data sourced from the National Health and Nutrition Examination Survey (NHANES) carried out from 2011 to 2014. The NHANES is conducted by the National Center for Health Statistics (NCHS) of the Centers for Disease Control and Prevention (CDC) in the United States. The NHANES survey collects data via trained experts from interviews and examinations to create a representative sample of the US population using a complex multistage, stratified, clustered probability sample. The survey protocol has been duly sanctioned by the Institutional Review Board of the CDC, and all participants have voluntarily given their informed consent.

Two cycles of dataset (2011–2012, 2013–2014) with full information were used for analysis. There was a total of 19,932 participants in the NHANES during 2011–2014. We have excluded individuals who failed to complete the survey or had missing or unreliable CMI data ($n = 14,054$) and PHQ-9 data ($n = 1265$). Besides, participants without complete information about BMI, education, marital status, PIR, smoking and drinking were also excluded. As a result, a total of 3794 participants have been included in our final analysis. The flowchart for participant screening was presented in Fig. 1.

2.2. Cardiometabolic index, lipid accumulation product (LAP), triglyceride glucose index (TyG-index) and visceral adiposity index (VAI) calculation, and depression definition

As previously described (Wakabayashi and Daimon, 2015), CMI was calculated based on the anthropometric data and biochemical data, including height, waist circumference (WC), triglyceride (TG), and high-density lipoprotein cholesterol (HDL-C). The units for height and WC were centimeters (cm) and for TG and HDL-C were milligrams per deciliter (mg/dl), respectively. The computational formula for CMI was presented as follows:

$$\text{WtHR} = \frac{\text{WC}}{\text{height}}$$

$$\text{CMI} = \frac{\text{TG}}{\text{HDL} - \text{C}} \times \text{WtHR}$$

Briefly, the WtHR was firstly calculated as the ratio of WC to height. Then, CMI was further calculated by multiplying the ratio of TG/HDL-C and WtHR.

LAP can be calculated by the following equation as previously reported (Ahn et al., 2019): $\text{LAP} = [\text{waist circumference (cm)} - 65] \times \text{TG (mmol/l)}$ (for males), or $\text{LAP} = [\text{waist circumference (cm)} - 58] \times \text{TG (mmol/l)}$ (for females). TyG-index can be got by the following equation (Ahn et al., 2019): $\text{TyG-index} = \text{Ln}[(\text{TG (mg/dl)} \times \text{glucose (mg/dl)})/2]$. Besides, VAI can be measured by the following formulas (Ahn et al., 2019): $\text{VAI} = \frac{\text{Waist circumference}}{39.68 + (1.88 \times \text{BMI})} \times \frac{\text{TG}}{1.03} \times \frac{1.31}{\text{HDL}}$ (for males), or $\text{VAI} = \frac{\text{Waist circumference}}{36.58 + (1.89 \times \text{BMI})} \times \frac{\text{TG}}{0.81} \times \frac{1.52}{\text{HDL}}$ (for females).

Depression was assessed using the Patient Health Questionnaire (PHQ-9), a self-report measure based on the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV). The PHQ-9

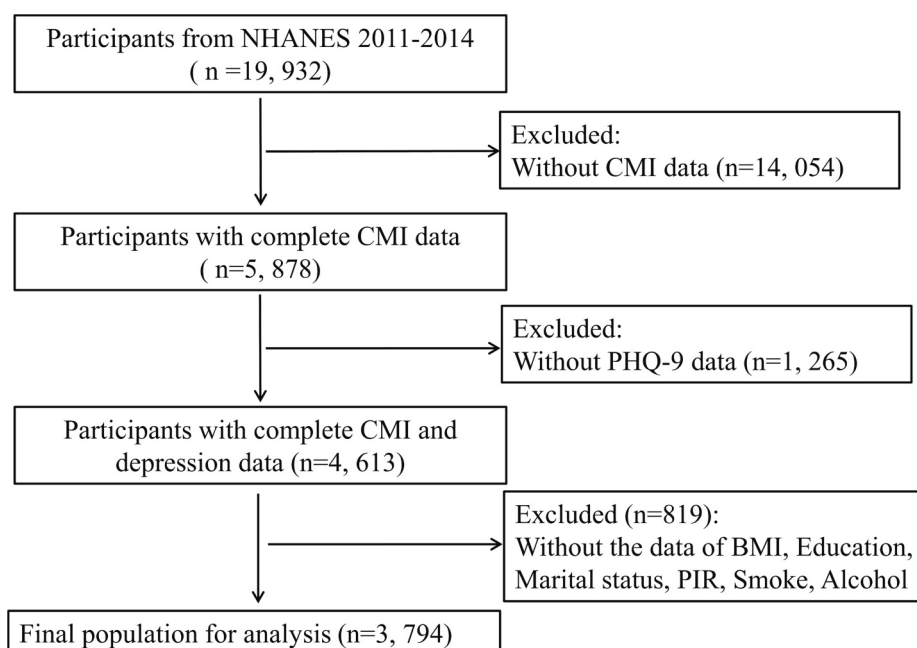


Fig. 1. Flow chart of participant selection.

consists of nine items that capture the signs and symptoms of depression experienced over the past two weeks. Participants rated each item on a 4-point scale ranging from 0 (not at all) to 3 (nearly every day) (Cai et al., 2022, pp. 2005–2018; Shi et al., 2021, pp. 2005–2018). The item scores were summed to obtain a total score ranging from 0 to 27 for each participant. Consistent with previous research, a cut-off score of ≥ 10 was used to define depression in this study (Dong et al., 2022). This threshold has been widely used in clinical and epidemiological studies and has been clinically validated with a sensitivity and specificity of 88 %.

2.3. Covariates

The covariates were selected based on previous literature and substantive reasoning that suggested their potential association with both CMI levels and depression prevalence. Standardized questionnaires were utilized to gather information on various factors, including age, gender, race, education level, marital status, poverty income ratio (PIR), body mass index (BMI), smoking status, alcohol use, and cardiovascular disease (CVD) history. Race was categorized into four groups: Mexican American, Non-Hispanic Black, Non-Hispanic White, and other races. PIR was stratified into three levels: <1.30 , 1.30 – 3.49 , and ≥ 3.50 . Marital status was categorized into three groups: married, unmarried, and others (widowed, divorced, and separated). Education was stratified into five levels: less than 9th grade, 9th–11th grade, high school, some college, and college or above. Participants were classified as past smokers or current smokers based on their responses to questions about smoking at least 100 cigarettes during their lifetime and whether they were currently smoking. Alcohol status was determined based on a question that “Had at least 12 alcohol drinks a year?”. CVD was assessed by a set of self-reported questionnaires. The Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation was used to diagnose chronic kidney disease (CKD) based on an estimated glomerular filtration rate (eGFR) of <60 mL/min per 1.73 m² or albumin-to-creatinine Ratio (ACR) ≥ 30 mg/g (Wang et al., 2016). Hypertension was defined by participants reporting a diagnosis, having an average systolic blood pressure (SBP) of ≥ 130 mmHg or an average diastolic blood pressure (DBP) of ≥ 80 mmHg, or taking antihypertensive medication (Peng et al., 2023, pp. 2011–2018). Diabetes can be diagnosed through physician

confirmation or by meeting specific criteria. These criteria include having a glycosylated hemoglobin (HbA1c) level >6.5 %, a fasting glucose level of 7.00 mmol/L or higher, a random blood glucose level of 11.10 mmol/L or higher, a two-hour oral glucose tolerance test (OGTT) result of 11.10 mmol/L or higher, or current use of diabetes medication or insulin (Ou et al., 2023, p. 2). Detailed information on the specimen collection, processing, quality assurance, and monitoring are described in the section of biospecimen program in NHANES.

2.4. Statistical analysis

All statistical analyses were conducted in accordance with the guidelines of the CDC, and sampling weights were used to produce nationally representative prevalence estimates for the non-institutionalized US population. Continuous variables were presented as mean $\pm$ standard deviation (SD), while categorical variables were reported as proportions. A weighted student's *t*-test (continuous variables) or a weighted chi-square test (categorical variables) was used for comparison of differences between the two groups. Weighted logistic regression model was used to analyze the association between CMI and depression. The performance of CMI and other indicators (including VAI, LAP and TrG-index) in identifying depression was compared by utilizing receiver operating curve (ROC) and calculating area under curve (AUC). Subgroup analysis and interaction tests were used to confirm the stability of the association between CMI and depression across different subgroups. Restricted cubic spline (RCS) regression was utilized to explore the nonlinear relationships between CMI and depression. The method of multiple imputation was utilized to deal with the variables with missing data. A two-sided $P < 0.05$ was considered statistically significant. Sample weighting was applied for the unequal probability of individual sampling, resulting from complex multistage survey design. The R software (version 4.3.1) was employed for all analyses, and statistical significance was ascertained by a two-sided P value <0.05 .

3. Results

3.1. Baseline characteristics of the study population

The baseline characteristics of all participants based on grouping of depression were presented in Table 1. The mean age of all enrolled subjects was 48.38 ± 17.53 years, with 50.47 % of them being female, and 44.23 % of them being non-Hispanic white. The weighted prevalence of depression was 8.51 % in the entire US population.

Compared with the non-depressive group, individuals in the depressive group exhibited a higher level of CMI (1.30 ± 1.10 vs. 1.04 ± 0.87), LAP (107.52 ± 87.32 vs. 56.43 ± 82.71), VAI (151.83 ± 170.08 vs. 100.69 ± 185.98), TyG-index (9.02 ± 0.75 vs. 8.50 ± 0.64), as well as higher levels of age and BMI. Besides, non-depressive participants were more likely to have low education levels, low family annual income, and be a current drinker, and have multiple chronic comorbidities, such as CVD, CKD, diabetes, hypertension, hyperlipidemia, and so on. However, there was no significant statistical difference in race between people with and without depression.

3.2. Association between cardiometabolic index (CMI) and depression

The weighted logistic regression analysis was utilized to explore the association between CMI and depression, and the results were presented in Table 2. A positive correlation between CMI (as continuous variable or categorical variable) and the occurrence of depression was found in all models (including the non-adjusted and adjusted models). When CMI was analyzed as continuous variable, each unit increase in CMI was correlated with a 34 %, 28 % and 34 % increase in odds ratio for the occurrence of depression from model 1 to model 3, respectively. After converting CMI into a categorical variable, the prevalence of depression also increased with increasing tertiles of CMI (P for trend <0.001 , in three models). In model 3, participants in the highest CMI tertile group were more liable to develop depression than those in the middle and low CMI tertile group (OR, 2.28 vs. 1.54 vs. 1, P for trend <0.001).

The associations between other cardiometabolic indicators and depression were also explored, including VAI, LAP and TyG-index (sTable 1). Similar to CMI, when LAP and TrG-index were analyzed as a continuous variable, each unit increase in CMI was significantly associated with higher likelihood of depression for LAP and TrG-index in all models. However, when LAP and TrG-index were converted to categorical variables (tertiles), the associations between LAP and TrG-index and depression became insignificant in the fully adjusted model, suggesting the instable associations between LAP and TrG-index and depression. Regarding VAI, although VAI was significantly associated with depression in model 1 and model 2 when VAI was analyzed as continuous variable or categorical variable, such association also became insignificant in the fully adjusted model (model 3). These results together revealed that VAI, LAP and TyG-index cannot get favorable performance in discriminating or predicting depression.

Additionally, the ability of CMI and other three cardiometabolic indicators to identify depression was further analyzed by using ROC curve (sFig. 1). Compared with other three indicators (with the AUC of 0.567, 0.552 and 0.516 for TrG-index, VAI and LAP, respectively), CMI got a significantly higher AUC of 0.748, suggesting an superior performance in identifying depression.

3.3. Analysis of restricted cubic spline (RCS) regression

In restricted cubic spline regression, after the adjustment of potential covariates, a significant nonlinear relationship between CMI and depression (P for nonlinearity <0.001) was detected (Fig. 2). A trend of increased OR for depression accompanying by increasing CMI was observed between them. RCS analysis identified two significant inflection points: the first at 0.9522 and the second at 1.58. Individuals with a CMI level below 0.9522 had a very low risk of developing depression.

Table 1
Baseline characteristics of the study participants.

Characteristics	Overall	Depression		P-value
		Without	With	
n	3794	3471	323	
Age, years	48.38 ± 17.53	48.10 ± 17.61	51.46 ± 16.26	<0.001
BMI, kg/m ²	28.90 ± 6.88	28.64 ± 6.71	31.60 ± 8.08	<0.001
Gender, n(%)				<0.001
Female	1915 (50.47 %)	1711 (49.29 %)	204 (63.16 %)	
Male	1879 (49.53 %)	1760 (50.71 %)	119 (36.84 %)	
Race, n(%)				0.202
Mexican	439 (11.57 %)	404 (11.64 %)	35 (10.84 %)	
American	771 (20.32 %)	708 (20.40 %)	63 (19.50 %)	
Non-Hispanic Black	1678 (44.23 %)	1518 (43.73 %)	160 (49.54 %)	
Non-Hispanic White	906 (23.88 %)	841 (24.23 %)	65 (20.12 %)	
Other races				
Education, n(%)				<0.001
Less than 9th grade	254 (6.69 %)	218 (6.28 %)	36 (11.15 %)	
9th–11th grade	512 (13.49 %)	449 (12.94 %)	63 (19.50 %)	
High school	789 (20.80 %)	710 (20.46 %)	79 (24.46 %)	
Some college	1185 (31.23 %)	1081 (31.14 %)	104 (32.20 %)	
College or above	1054 (27.78 %)	1013 (29.18 %)	41 (12.69 %)	
Marital Status, n(%)				<0.001
Unmarried	774 (20.40 %)	721 (20.77 %)	53 (16.41 %)	
Married	1958 (51.61 %)	1840 (53.01 %)	118 (36.53 %)	
Others	1062 (27.99 %)	910 (26.22 %)	152 (47.06 %)	
PIR, n(%)				<0.001
<1.30	1259 (33.18 %)	1073 (30.91 %)	186 (57.59 %)	
1.30–3.49	1324 (34.90 %)	1231 (35.47 %)	93 (28.79 %)	
≥3.50	1211 (31.92 %)	1167 (33.62 %)	44 (13.62 %)	
Smoke, n(%)				<0.001
Never Smoker	1008 (26.57 %)	952 (27.43 %)	56 (17.34 %)	
Former Smoker	2030 (53.51 %)	1871 (53.90 %)	159 (49.23 %)	
Current Smoker	756 (19.93 %)	648 (18.67 %)	108 (33.44 %)	
Alcohol Use, n(%)				0.003
Never drinker	546 (14.39 %)	500 (14.41 %)	46 (14.24 %)	
Former drinker	449 (11.83 %)	392 (11.29 %)	57 (17.65 %)	
Current drinker	2799 (73.77 %)	2579 (74.30 %)	220 (68.11 %)	
CVD, n(%)				<0.001
Yes	274 (7.22 %)	225 (6.48 %)	49 (15.17 %)	
No	3520 (92.78 %)	3246 (93.52 %)	274 (84.83 %)	
CKD, n(%)				0.003
Yes	383 (10.09 %)	335 (9.65 %)	48 (14.86 %)	
No	3411 (89.91 %)	3136 (90.35 %)	275 (85.14 %)	
Hypertension, n(%)				<0.001
Yes	1969 (51.90 %)	1761 (50.73 %)	208 (64.40 %)	
No	1825 (48.10 %)	1710 (49.27 %)	115 (35.60 %)	
Hyperlipidemia, n(%)				<0.001

(continued on next page)

Table 1 (continued)

Characteristics	Overall	Depression		P-value
		Without	With	
Yes	2067 (54.48 %)	1849 (53.27 %)	218 (67.49 %)	
No	1727 (45.52 %)	1622 (46.73 %)	105 (32.51 %)	
Diabetes, n(%)				<0.001
Yes	711 (18.74 %)	618 (17.80 %)	93 (28.79 %)	
No	3083 (81.26 %)	2853 (82.20 %)	230 (71.21 %)	
CMI	1.06 ± 0.87	1.04 ± 0.87	1.30 ± 1.10	<0.001
LAP	63.07 ± 85.07	56.43 ± 82.71	107.52 ± 87.32	<0.001
VAI	108.27 ± 184.57	100.69 ± 185.98	151.83 ± 170.08	<0.001
TyG-index	8.57 ± 0.68	8.50 ± 0.64	9.02 ± 0.75	<0.001

Abbreviations: CMI, cardiometabolic index; BMI, body mass index; PIR, poverty income ratio; CVD, cardiovascular disease; CKD, chronic kidney disease.

However, those with a CMI level between 0.9522 and 1.58 had a significantly higher risk of experiencing depression. Beyond a CMI level of 1.58, there was a gradual increase in the risk of depression. Therefore, it is important to maintain a CMI level below 0.9522 in order to potentially reduce the risk of developing depression.

In order to identify the stability of the association between CMI and depression, the whole population was divided into three subgroups based on the inflection points of CMI identified from RCS analysis, including ≤ 0.9522 , $0.9522\text{--}1.58$, and ≥ 1.58 . The weighted logistic regression analysis was utilized to explore the association between CMI (as categorical variable) and depression, and the results were presented in Table 3. Compared with individuals with a CMI lower than 0.9522, individuals with a CMI level between 0.9522 and 1.58, or with a CMI level higher than 1.58 both showed a significant increased risk for the

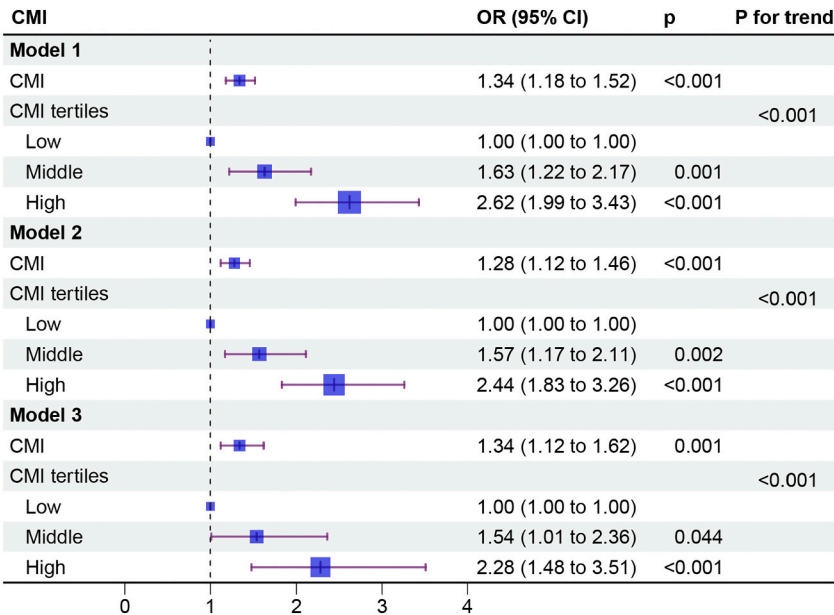
development of depression in non-adjusted and adjusted models (all P value <0.05 , and all P for trend <0.05).

CMI was initially proposed by Ichiro Wakabayashi et al. as a excellent predictor for coronary artery disease, metabolic syndrome, hyperglycemia and diabetes (Wakabayashi and Daimon, 2015). The cut-off values of CMI for hyperglycemia and diabetes were determined to be 0.799 and 0.800 in female, respectively, and to be 1.625 and 1.748 in male, respectively. In order to determine whether the cut-off values for hyperglycemia (CMI_Hyperglycemia) and diabetes (CMI_Diabetes) can serve as reference values to discriminate depression, we divided the entire group into two groups based the cut-off values for hyperglycemia and diabetes. The results from multivariate logistic regression analysis revealed that higher level of CMI was significantly associated with higher risk of depression in model 1 and model 2, but such associations became insignificant in the fully adjusted model (sTable 2). Additionally, the ROC curve analysis also indicated that the AUC for CMI_Hyperglycemia and CMI_Diabetes was 0.640 and 0.625, respectively, which was significantly lower than that of the CMI_RCS (sFig. 2).

3.4. Association between CMI and depression in different subgroups

The subgroup analyses were conducted in order to scrutinize the reliability and robustness of the relationship between CMI and depression across different subgroups. The summarized results for subgroup analyses were presented in Table 4. Consistent with the findings obtained from previous analyses in the whole population, a higher level of CMI was associated with increased OR for depression, which demonstrated consistent outcomes across all subgroups. Besides, interaction tests revealed that such association between CMI and depression was not modified by other covariates except for the variable of drinking (all P for interaction >0.05), which was marginally significant. When the former drinkers and never drinkers were categorized into one group (namely nondrinkers), interaction tests suggested that the association between CMI and depression was significantly modified by the variable of

Table 2
Associations between CMI and depression.



Note: CMI was converted from a continuous variable to a categorical variable (tertiles). **Abbreviations:** CMI, cardiometabolic index; CI, confidence interval; OR, odds ratio.
Model 1: no covariates were adjusted.
Model 2: adjusted for age, gender and race.
Model 3: adjusted for age, gender, race, education level, marital status, PIR, BMI, smoke, alcohol use, CVD, CKD, hypertension, hyperlipidemia, and diabetes.

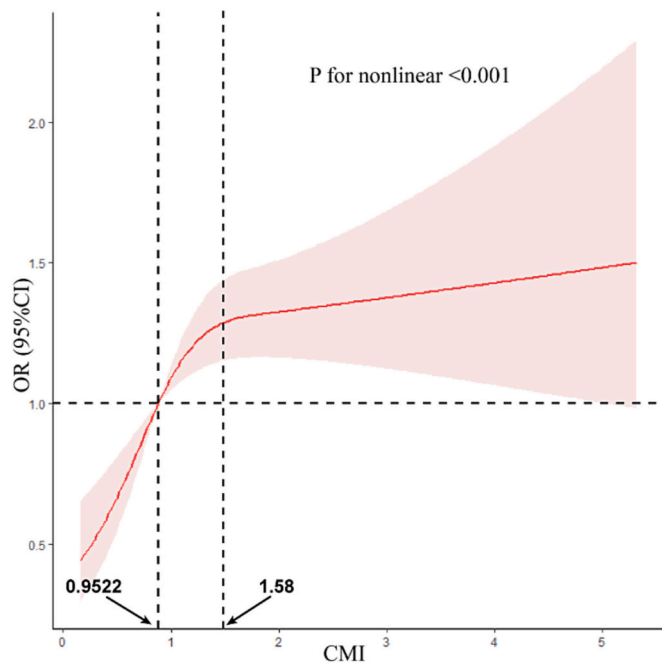


Fig. 2. Determination of the association between CMI and depression by restricted cubic spline (RCS) regression analysis.

Table 3
Association between CMI (as a three categorical variable) and depression based on the inflection points identified from RCS analysis.

CMI categorical	Model 1		Model 2		Model 3	
	OR (95 % CI)	P-value	OR (95 % CI)	P-value	OR (95 % CI)	P-value
≤0.9522	1		1		1	
0.9522–1.58	2.20 (1.73, 2.78)	<0.001	2.10 (1.64, 2.68)	<0.001	1.87 (1.27, 2.76)	0.0015
≥1.58	2.82 (2.15, 3.70)	<0.001	2.67 (2.01, 3.54)	<0.001	2.05 (1.27, 3.30)	0.0033
P for trend	<0.001		<0.001		0.002	

Note: CMI was converted from a continuous variable to a three categorical variable based on the inflection points identified from restricted cubic spline (RCS) regression analysis. **Abbreviations:** CMI, cardiometabolic index; CI, confidence interval; OR, odds ratio. Model 1: no covariates were adjusted; Model 2: adjusted for age, gender and race; Model 3: adjusted for age, gender, race, education level, marital status, PIR, BMI, smoke, alcohol use, CVD, CKD, hypertension, hyperlipidemia, and diabetes.

drinking. Significant positive association between CMI and depression was observed in the subgroup of current drinkers, but not in the subgroup of former or never drinkers (*sTable 3*). Finally, RCS analysis also revealed that a higher CMI level in current drinkers got obviously higher OR for suffering depression compared with former or never drinkers (*sFig. 3*). In contrast, such phenomena was not observed between different genders (*sFig. 4*).

4. Discussion

To our knowledge, this study was the first to examine the association between CMI and depression. The results showed that higher CMI levels were associated with an increased risk of developing depression, regardless of whether CMI was analyzed as a continuous or categorical variable. The relationship between CMI and depression was found to be non-linear, with two significant inflection points identified at 0.9522

and 1.58, respectively. Logistic regression analysis confirmed a positive correlation between higher CMI and depression when CMI was analyzed as a three categorical variable (divided by the inflection points of 0.9522 and 1.58). Subgroup analyses and interaction tests revealed that the association between CMI and depression was consistent across different subgroups and not influenced by other covariates except for drinking. The positive correlation between CMI and depression was more pronounced in those with who current drinkers.

The metabolic profile that leads to depression has attracted the attention of scientists in recent years (Catoira et al., 2016). As an indicator for evaluating metabolic profile by assessing both abdominal obesity and blood lipid (Miao et al., 2023), our results observed a robust association between higher levels of CMI and increased occurrence of depression. There are several reasons that may account for such strong association. Firstly, the inflammatory response caused by abdominal obesity and increased blood lipids may leads to such association (Guo et al., 2023). Amounting evidences have suggested that abdominal obesity and increased blood lipids are often accompanied by an increase in chronic inflammation, which may affect the nervous system and emotional regulation, and thus leads to depressive disorders (Liu et al., 2022; Shelton and Miller, 2010). Secondly, metabolic disorders usually result in the imbalance of the neuroendocrine system (Kivimäki et al., 2023), such as insulin resistance, increased secretion of glucocorticoid, and disturbance of glucocorticoid sensitivity associated with systemic cortisol activity (Jakobsen et al., 2020; Steptoe and Frank, 2023). These changes may interfere with neurogenesis and neural plasticity, and thus cause depression. Thirdly, the hormonal alterations were not only associated with metabolic changes, but also can impact the neural inputs of the reward circuit, which may ultimately result in the manifestation of anhedonia and thus depression (Bangasser and Cuarenta, 2021). Besides, since high CMI levels were reported to usually coexist with various additional risk factors, such as oxidative stress, self-esteem, social support, and the capacity to modify self-efficacy, and so on (Koborová et al., 2018; Monserrat-Mesquida et al., 2023; Świątkiewicz et al., 2023). Therefore, these risk factors may collectively contribute to an elevated susceptibility to developing depression (Kandola et al., 2019). Given that the strong association between CMI and depression, in clinical contexts, it is crucial to conduct a thorough assessment of individuals with elevated CMI levels and to adopt efficacious interventions for mitigating risk factors, thereby reducing the likelihood of depression.

The present study revealed a significant association between higher CMI levels and an increased prevalence of depression, even after adjusting for potential confounding variables. This suggests that a more severe disturbance in metabolic profile is correlated with a higher incidence of depression. Interestingly, this association between CMI levels and depression prevalence remained consistent across most subgroups. Notably, individuals categorized as obese or overweight (with a BMI ≥ 25 kg/m²) showed a robust association between CMI levels and depression prevalence, unlike their non-obese or non-overweight counterparts. Surprisingly, individuals with hyperlipidemia had a lower depression prevalence and lower CMI levels compared to the population without hyperlipidemia. However, within the hyperlipidemia group, a higher level of CMI was associated with a higher incidence of depression. These findings highlight the importance of considering multiple factors, including unhealthy lifestyle habits and biological changes, when addressing abdominal obesity, hyperlipidemia, and depression.

In this study, the interaction by alcohol use was marginally significant when the drinking status was divided into three subgroups, including never drinker, former drinker and current drinker. When the never drinkers and former drinkers were categorized into one group (namely as nondrinker group), interaction tests revealed that the association between CMI and depression was significantly modified by drinking status. Actually, drinking has already been reported to be closely associated with CMI and depression in numerous previous studies. Their relationship can be illuminated from the following

Table 4

Verification of the association between CMI and depression by subgroup analyses.

Subgroup		OR (95% CI)	P value	P for interaction
Age, years				0.666
<60		2.49 (1.92 to 3.23)	<0.001	
≥60		2.21 (1.51 to 3.25)	<0.001	
Gender				0.436
Female		2.17 (1.65 to 2.85)	<0.001	
Male		2.63 (1.87 to 3.71)	<0.001	
Race				0.6295
Mexican American		2.56 (1.29 to 5.07)	0.007	
Non-Hispanic Black		1.75 (1.11 to 2.77)	0.017	
Non-Hispanic White		3.01 (2.17 to 4.18)	<0.001	
Other races		2.13 (1.34 to 3.39)	0.001	
BMI, kg/m ²				0.8644
<25		1.82 (1.18 to 2.82)	0.007	
≥25		2.53 (1.94 to 3.29)	<0.001	
Education				0.4385
Less than 9th grade		2.33 (1.20 to 4.50)	0.012	
9th–11th grade		2.80 (1.68 to 4.66)	<0.001	
High school		2.01 (1.29 to 3.14)	0.002	
Some college		2.08 (1.39 to 3.10)	<0.001	
College or above		4.29 (2.24 to 8.23)	<0.001	
Marital Status				0.1867
Unmarried		2.56 (1.54 to 4.25)	<0.001	
Married		2.85 (1.94 to 4.18)	<0.001	
Others		2.25 (1.61 to 3.14)	<0.001	
PIR				0.2848
< 1.30		2.50 (1.85 to 3.37)	<0.001	
1.30–3.49		2.63 (1.73 to 4.00)	<0.001	
≥3.50		2.25 (1.25 to 4.02)	0.007	
Smoke				0.193
Never Smoker		2.82 (1.68 to 4.72)	<0.001	
Former Smoker		2.24 (1.66 to 3.03)	<0.001	
Current Smoker		2.60 (1.74 to 3.87)	<0.001	
Alcohol Use				0.0525
Never drinker		2.70 (1.57 to 4.65)	<0.001	
Former drinker		3.02 (1.65 to 5.54)	<0.001	
Current drinker		2.09 (1.60 to 2.72)	<0.001	
CVD				0.5985
Yes		2.43 (1.29 to 4.57)	0.006	
No		2.51 (1.97 to 3.18)	<0.001	
CKD				0.6625
Yes		2.48 (1.34 to 4.59)	0.004	
No		2.41 (1.77 to 3.28)	<0.001	
Hypertension				0.5212
Yes		2.29 (1.72 to 3.05)	<0.001	
No		2.14 (1.52 to 2.99)	<0.001	
Hyperlipidemia				0.998
Yes		2.12 (1.59 to 2.83)	<0.001	
No		2.30 (1.61 to 3.29)	<0.001	
Diabetes				0.5187
Yes		2.61 (1.59 to 4.30)	<0.001	
No		2.19 (1.71 to 2.79)	<0.001	

Abbreviations: CMI, cardiometabolic index; BMI, body mass index; PIR, poverty income ratio; CVD, cardiovascular disease; CKD, chronic kidney disease. CMI was analyzed as continuous variable. Variables of age, gender, race, education level, marital status, PIR, BMI, smoke, alcohol use, CVD, CKD, hypertension, hyperlipidemia, and diabetes were adjusted in subgroup analyses.

aspects. Firstly, regarding the relationship between drinking and CMI, a U-shaped relationship between alcohol consumption and cardiometabolic index in middle-aged men was reported by Ichiro Wakabayashi (Wakabayashi, 2016), the log-transformed CMI was significantly lower in light (being lowest in all drinker groups) and moderate drinkers compared with nondrinkers and very heavy drinkers. Therefore, the drinking pattern may play a vital role in strengthening or attenuating the relation of CMI with other variables. In our study, we found that the relationship between CMI and depression was more pronounced in subjects who are current drinkers, but not in never or former drinkers (sFig. 3). The RCS curve reveals that the OR for depression increases significantly as CMI continues to elevate in the subgroup of current drinkers. However, in the subgroup of never and former drinkers group, the OR for depression decreases smoothly as CMI exceeds the value of approximate 1.5, a point at which subjects get the highest OR for depression. In our study, since the drinking status was got from the Alcohol Use Questionnaire (variable name prefix ALQ), which was focusing on the lifetime and current use (past 12 months) of alcohol, and the questions are not specific to type of alcohol used, it is very difficult to define the detail status of drinking as reported in previous studies. Secondly, previous studies have reported that increasing involvement with alcohol significantly increases the risk of depression (Boden and Fergusson, 2011; Zhang et al., 2021), and individuals with depressive symptoms or anxiety symptoms are more likely to have unhealthy patterns of alcohol drinking (Johannessen et al., 2017), suggesting that drinking and depression can affect each other. Therefore, alcohol drinking is a vital variable which may influence the relationship between CMI and depression, which should be taken into account when CMI is used as an index related to the risk of depression. Further studies are still needed to elucidate the detailed relationship between drinking, CMI and depression in the future.

The current study has several key strengths that contribute to the reliability and significance of the findings. Firstly, this study is the first to investigate the association between CMI levels and depression prevalence, providing novel insights into the relationship between metabolic profile and mental health. Secondly, the use of NHANES data, which employs a rigorous multistage probability sampling design, ensures that the results are representative of the broader U.S. population. This enhances the generalizability and applicability of the findings. Thirdly, the study has taken into account potential confounding factors, allowing for a more accurate assessment of the association between CMI levels and depression prevalence. Consequently, the results have important implications for public health strategies aimed at preventing depression.

There are several limitations in our study that should be acknowledged. Firstly, the study sample was limited to specific demographic areas, which may limit the generalizability of the findings to other populations. Conducting large-scale studies that include a more diverse range of regions and populations would enhance the external validity of the results. Secondly, the sample size in our study was relatively small, which may affect the reliability and generalizability of the findings. Replicating the study with a larger sample size would provide more robust evidence. Additionally, there is a need to standardize and compare the various methods used to assess metabolic profile to ensure consistency and comparability of results across studies. Thirdly, the cross-sectional design of our study limits our ability to establish causality between CMI levels and depression prevalence. Conducting long-term randomized controlled trials would provide more definitive evidence for establishing a causal relationship between CMI levels and depression prevalence. Collectively, while our study provides valuable insights, these limitations should be taken into account when interpreting the results.

5. Conclusion

In conclusion, our investigation utilizing a nationally representative cohort of US population revealed a robust positive correlation between

CMI and depression. A positive, and non-linear association between CMI and depression was observed by RCS analysis, with two significant inflection points being identified, including 0.9522 and 1.58. Individuals with a higher level of CMI is significantly associated with increased risk for developing depression. These findings suggest that CMI can serve as a useful indicator for assessing the risk of depression in individuals with dyslipidemia. Besides, maintaining a CMI level below 0.9522 may help to reduce the risk of developing depression.

CRedit authorship contribution statement

Xiang Zhou: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xiao-Liang Tao:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Writing – review & editing. **Li Zhang:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Qian-Kun Yang:** Conceptualization, Formal analysis, Methodology, Software, Validation, Visualization, Writing – review & editing. **Zi-Jiao Li:** Conceptualization, Data curation, Formal analysis, Investigation, Resources, Writing – review & editing. **Lu Dai:** Data curation, Investigation, Resources, Validation, Visualization. **Ya Lei:** Data curation, Investigation, Resources, Validation, Visualization. **Gang Zhu:** Data curation, Formal analysis, Investigation, Methodology, Resources. **Zhi-Feng Wu:** Data curation, Investigation, Methodology, Software, Validation. **Hui Yang:** Methodology, Software, Validation. **Kai-Feng Shen:** Data curation, Funding acquisition, Investigation, Methodology, Software, Validation. **Chun-Mei Xu:** Data curation, Investigation, Methodology, Software, Validation. **Ping Liang:** Methodology, Project administration, Supervision. **Xin Zheng:** Conceptualization, Project administration, Supervision.

Ethics approval and consent to participate

The National Health and Nutrition Examination Survey (NHANES) is a publicly available database and approved by the National Center for Health Statistics institutional review board. All participants provide written informed consent when they did the national survey in the United States. Ethical review and approval were waived for this study since secondary analysis did not require additional institutional review board approval.

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Declaration of competing interest

None.

Data availability

Data used for this study are available on the NHANES website: <https://www.cdc.gov/nchs/nhanes/>.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2024.02.024>.

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